Experiment 5

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mahin-mr12 commit 9eed2b6 · 3 weeks ago

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Preview Code Blame Raw

Experiment 5: String and Sets

1. Write a program to count and display the number of capital letters in a given

string.

In [1]:

```
text = "Hello World PYTHON"

count = 0
for ch in text:
    if ch.isupper():
        count += 1

print("Capital letters:", count)
```

```
Capital letters: 8
```

2. Count total number of vowels in a given string.

In [2]:

```
text = "Programming is fun"
vowels = "aeiouAEIOU"
count = 0

for ch in text:
    if ch in vowels:
        count += 1

print("Vowels:", count)
```

```
Vowels: 5
```

3. Input a sentence and print words in separate lines.

In [3]:

```
sentence = "Python is very easy"

words = sentence.split()
```

```python
for word in words:
    print(word)
```

```
Python
is
very
easy
```

4. WAP to enter a string and a substring. You have to print the number of times that the substring occurs in the given string. String traversal will take place from left to right, not from right to left.Sample

Input ABCDCDC

CDC

Sample Output

2

In [4]:

```python
string = "ABCDCDC"
sub = "CDC"

count = 0
for i in range(len(string) - len(sub) + 1):
    if string[i:i+len(sub)] == sub:
        count += 1

print(count)
```

```
2
```

5. Given a string containing both upper and lower-case alphabets. Write a Python program to count the number of occurrences of each alphabet (case insensitive) and display the same.

Sample Input

ABaBCbGc

Sample Output

2A

3B

2C

1G

In [5]:

```
text = "ABaBCbGc"
text = text.upper()

freq = {}

for ch in text:
    if ch.isalpha():
        freq[ch] = freq.get(ch, 0) + 1

for key in sorted(freq):
    print(freq[key], key)
```

```
2 A
3 B
2 C
1 G
```

6. Program to count the number of unique words in a given sentence using sets.

In [6]:

```
sentence = "this is python and python is easy"

words = sentence.split()
unique_words = set(words)

print("Unique words:", len(unique_words))
```

```
Unique words: 5
```

7. Create 2 sets s1 and s2 of n fruits each by taking input from user and find:

a. Fruits which are in both sets s1 and s2

b. Fruits only in s1 but not in s2

c. Count of all fruits from s1 and s2

In [7]:

```
s1 = {"apple", "banana", "mango", "grapes"}
```

```
s2 = {"banana", "grapes", "kiwi", "orange"}

print("Common fruits:", s1 & s2)
print("Only in s1:", s1 - s2)
print("Total fruits:", len(s1 | s2))
```

```
Common fruits: {'grapes', 'banana'}
Only in s1: {'mango', 'apple'}
Total fruits: 6
```

8. Take two sets and apply various set operations on them:

S1 = {Red, yellow, orange, blue}

S2 = {violet, blue, purple}

In [8]:

```
S1 = {"Red", "yellow", "orange", "blue"}
S2 = {"violet", "blue", "purple"}

print("Union:", S1 | S2)
print("Intersection:", S1 & S2)
print("Only S1:", S1 - S2)
print("Only S2:", S2 - S1)
print("Symmetric difference:", S1 ^ S2)
```

```
Union: {'purple', 'yellow', 'orange', 'Red', 'blue', 'violet'}
Intersection: {'blue'}
Only S1: {'Red', 'yellow', 'orange'}
Only S2: {'purple', 'violet'}
Symmetric difference: {'purple', 'violet', 'yellow', 'orange', 'Red'}
```